13. Let $\langle X,Y\rangle$ denote the set of basepoint-preserving homotopy classes of basepoint-preserving maps $X\to Y$. Using Proposition 1B.9, show that if X is a connected CW complex and G is an abelian group, then the map $\langle X,K(G,1)\rangle\to H^1(X;G)$ sending a map $f:X\to K(G,1)$ to the induced homomorphism $f_*:H_1(X)\to H_1(K(G,1))\approx G$ is a bijection, where we identify $H^1(X;G)$ with $\mathrm{Hom}(H_1(X),G)$ via the universal coefficient theorem.

3.2 Cup Product

In the introduction to this chapter we sketched a definition of cup product in terms of another product called cross product. However, to define the cross product from scratch takes some work, so we will proceed in the opposite order, first giving an elementary definition of cup product by an explicit formula with simplices, then afterwards defining cross product in terms of cup product. The other approach of defining cup product via cross product is explained at the end of §3.B.

To define the cup product we consider cohomology with coefficients in a ring R, the most common choices being $\mathbb{Z}$, $\mathbb{Z}_n$, and $\mathbb{Q}$. For cochains $\varphi\in C^k(X;R)$ and $\psi\in C^\ell(X;R)$, the **cup product** $\varphi\smile\psi\in C^{k+\ell}(X;R)$ is the cochain whose value on a singular simplex $\sigma:\Delta^{k+\ell}\to X$ is given by the formula

$$(\varphi\smile\psi)(\sigma)=\varphi(\sigma\,|\,[v_0,\cdots,v_k])\psi(\sigma\,|\,[v_k,\cdots,v_{k+\ell}])$$

where the right-hand side is the product in R. To see that this cup product of cochains induces a cup product of cohomology classes we need a formula relating it to the coboundary map:

Lemma 3.6. *$\delta(\varphi\smile\psi)=\delta\varphi\smile\psi+(-1)^k\varphi\smile\delta\psi$ for $\varphi\in C^k(X;R)$ and $\psi\in C^\ell(X;R)$.*

Proof: For $\sigma:\Delta^{k+\ell+1}\to X$ we have

$$(\delta\varphi\smile\psi)(\sigma)=\sum_{i=0}^{k+1}(-1)^i\varphi(\sigma|[v_0,\cdots,\hat v_i,\cdots,v_{k+1}])\psi(\sigma|[v_{k+1},\cdots,v_{k+\ell+1}])$$
$$(-1)^k(\varphi\smile\delta\psi)(\sigma)=\sum_{i=k}^{k+\ell+1}(-1)^i\varphi(\sigma|[v_0,\cdots,v_k])\psi(\sigma|[v_k,\cdots,\hat v_i,\cdots,v_{k+\ell+1}])$$

When we add these two expressions, the last term of the first sum cancels the first term of the second sum, and the remaining terms are exactly $\delta(\varphi\smile\psi)(\sigma)=(\varphi\smile\psi)(\partial\sigma)$ since $\partial\sigma=\sum_{i=0}^{k+\ell+1}(-1)^i\sigma\,|\,[v_0,\cdots,\hat v_i,\cdots,v_{k+\ell+1}]$. $\square$

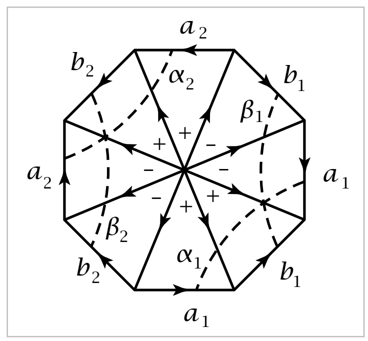
From the formula $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi \pm \varphi \smile \delta\psi$ it is apparent that the cup product of two cocycles is again a cocycle. Also, the cup product of a cocycle and a coboundary, in either order, is a coboundary since $\varphi \smile \delta\psi = \pm\delta(\varphi \smile \psi)$ if $\delta\varphi = 0$, and $\delta\varphi \smile \psi = \delta(\varphi \smile \psi)$ if $\delta\psi = 0$. It follows that there is an induced cup product

$$H^k(X;R) \times H^\ell(X;R) \xrightarrow{\smile} H^{k+\ell}(X;R)$$

This is associative and distributive since at the level of cochains the cup product obviously has these properties. If R has an identity element, then there is an identity element for cup product, the class $1 \in H^0(X;R)$ defined by the 0-cocycle taking the value 1 on each singular 0-simplex.

A cup product for simplicial cohomology can be defined by the same formula as for singular cohomology, so the canonical isomorphism between simplicial and singular cohomology respects cup products. Here are three examples of direct calculations of cup products using simplicial cohomology.

Example 3.7. Let M be the closed orientable surface of genus $g \geq 1$ with the Δ-complex structure shown in the figure for the case $g = 2$. The cup product of interest is $H^1(M) \times H^1(M) \to H^2(M)$. Taking $\mathbb{Z}$ coefficients, a basis for $H_1(M)$ is formed by the edges a_i and b_i, as we showed in Example 2.36 when we computed the homology of M using cellular homology. We have $H^1(M) \approx \mathrm{Hom}(H_1(M), \mathbb{Z})$ by cellular cohomology or the universal coefficient theorem. A basis for $H_1(M)$ determines a dual basis for $\mathrm{Hom}(H_1(M), \mathbb{Z})$, so dual to a_i is the cohomology class α_i assigning the value 1 to a_i and 0 to the other basis elements, and similarly we have cohomology classes β_i dual to b_i.



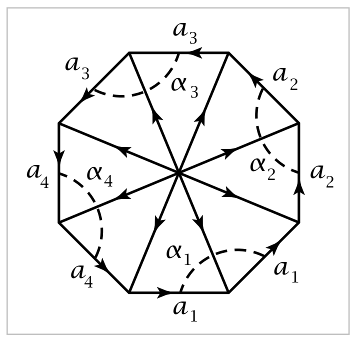
To represent α_i by a simplicial cocycle φ_i we need to choose values for φ_i on the edges radiating out from the central vertex in such a way that $\delta\varphi_i = 0$. This is the 'cocycle condition' discussed in the introduction to this chapter, where we saw that it has a geometric interpretation in terms of curves transverse to the edges of M. With this interpretation in mind, consider the arc labeled α_i in the figure, which represents a loop in M meeting a_i in one point and disjoint from all the other basis elements a_j and b_j. We define φ_i to have the value 1 on edges meeting the arc α_i and the value 0 on all other edges. Thus φ_i counts the number of intersections of each edge with the arc α_i. In similar fashion we obtain a cocycle ψ_i counting intersections with the arc β_i, and ψ_i represents the cohomology class β_i dual to b_i.

Now we can compute cup products by applying the definition. Keeping in mind that the ordering of the vertices of each 2-simplex is compatible with the indicated orientations of its edges, we see for example that $\varphi_1 \smile \psi_1$ takes the value 0 on all 2-simplices except the one with outer edge b_1 in the lower right part of the figure, where it takes the value 1. Thus $\varphi_1 \smile \psi_1$ takes the value 1 on the 2-chain c formed by the sum of all the 2-simplices with the signs indicated in the center of the figure. It is an easy calculation that $\partial c = 0$. Since there are no 3-simplices, c is not a boundary, so it represents a nonzero element of $H_2(M)$. The fact that $(\varphi_1 \smile \psi_1)(c)$ is a generator of $\mathbb{Z}$ implies both that c represents a generator of $H_2(M) \approx \mathbb{Z}$ and that $\varphi_1 \smile \psi_1$ represents the dual generator γ of $H^2(M) \approx \operatorname{Hom}(H_2(M), \mathbb{Z}) \approx \mathbb{Z}$. Thus $\alpha_1 \smile \beta_1 = \gamma$. In similar fashion one computes:

$$\alpha_i \smile \beta_j = \left\{ \begin{matrix} \gamma, & i = j \\ 0, & i \neq j \end{matrix} \right\} = -(\beta_i \smile \alpha_j), \qquad \alpha_i \smile \alpha_j = 0, \qquad \beta_i \smile \beta_j = 0$$

These relations determine the cup product $H^1(M) \times H^1(M) \to H^2(M)$ completely since cup product is distributive. Notice that cup product is not commutative in this example since $\alpha_i \smile \beta_i = -(\beta_i \smile \alpha_i)$. We will show in Theorem 3.14 below that this is the worst that can happen: Cup product is commutative up to a sign depending only on dimension.

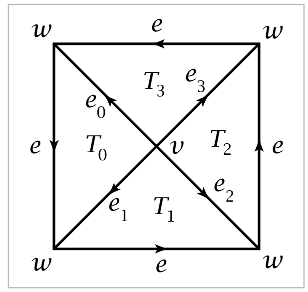
One can see in this example that nonzero cup products of distinct classes α_i or β_j occur precisely when the corresponding loops α_i or β_j intersect. This is also true for the cup product of α_i or β_i with itself if we allow ourselves to take two copies of the corresponding loop and deform one of them to be disjoint from the other.



Example 3.8. The closed nonorientable surface N of genus g can be treated in similar fashion if we use $\mathbb{Z}_2$ coefficients. Using the Δ-complex structure shown, the edges a_i give a basis for $H_1(N;\mathbb{Z}_2)$, and the dual basis elements $\alpha_i \in H^1(N;\mathbb{Z}_2)$ can be represented by cocycles with values given by counting intersections with the arcs labeled α_i in the figure. Then one computes that $\alpha_i \smile \alpha_i$ is the nonzero element of $H^2(N;\mathbb{Z}_2) \approx \mathbb{Z}_2$ and $\alpha_i \smile \alpha_j = 0$ for $i \neq j$. In particular, when $g = 1$ we have $N = \mathbb{R}\mathrm{P}^2$, and the cup product of a generator of $H^1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}_2)$ with itself is a generator of $H^2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}_2)$.

The remarks in the paragraph preceding this example apply here also, but with the following difference: When one tries to deform a second copy of the loop α_i in the present example to be disjoint from the original copy, the best one can do is make it intersect the original in one point. This reflects the fact that $\alpha_i \smile \alpha_i$ is now nonzero.

Example 3.9. Let X be the 2-dimensional CW complex obtained by attaching a 2-cell to S^1 by the degree m map $S^1 \to S^1$, $z \mapsto z^m$. Using cellular cohomology, or cellular homology and the universal coefficient theorem, we see that $H^n(X;\mathbb{Z})$ consists of a $\mathbb{Z}$ for $n = 0$ and a $\mathbb{Z}_m$ for $n = 2$, so the cup product structure with $\mathbb{Z}$ coefficients is uninteresting. However, with $\mathbb{Z}_m$ coefficients we have $H^i(X;\mathbb{Z}_m) \approx \mathbb{Z}_m$ for $i = 0, 1, 2$, so there is the possibility that the cup product of two 1-dimensional classes can be nontrivial.



To obtain a Δ-complex structure on X, take a regular m-gon subdivided into m triangles T_i around a central vertex v, as shown in the figure for the case $m = 4$, then identify all the outer edges by rotations of the m-gon. This gives X a Δ-complex structure with 2 vertices, $m+1$ edges, and m 2-simplices. A generator α of $H^1(X;\mathbb{Z}_m)$ is represented by a cocycle φ assigning the value 1 to the edge e, which generates $H_1(X)$. The condition that φ be a cocycle means that $\varphi(e_i) + \varphi(e) = \varphi(e_{i+1})$ for all i, subscripts being taken mod m. So we may take $\varphi(e_i) = i \in \mathbb{Z}_m$. Hence $(\varphi \smile \varphi)(T_i) = \varphi(e_i)\varphi(e) = i$. The map $h : H^2(X;\mathbb{Z}_m) \to \mathrm{Hom}(H_2(X;\mathbb{Z}_m),\mathbb{Z}_m)$ is an isomorphism since $\sum_i T_i$ is a generator of $H_2(X;\mathbb{Z}_m)$ and there are 2-cocycles taking the value 1 on $\sum_i T_i$, for example the cocycle taking the value 1 on one T_i and 0 on all the others. The cocycle $\varphi \smile \varphi$ takes the value $0+1+\cdots+(m-1)$ on $\sum_i T_i$, hence represents $0+1+\cdots+(m-1)$ times a generator β of $H^2(X;\mathbb{Z}_m)$. In $\mathbb{Z}_m$ the sum $0+1+\cdots+(m-1)$ is 0 if m is odd and k if $m = 2k$ since the terms 1 and $m-1$ cancel, 2 and $m-2$ cancel, and so on. Thus, writing α^2 for $\alpha \smile \alpha$, we have $\alpha^2 = 0$ if m is odd and $\alpha^2 = k\beta$ if $m = 2k$.

In particular, if $m = 2$, X is $\mathbb{R}\mathrm{P}^2$ and $\alpha^2 = \beta$ in $H^2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}_2)$, as we showed already in Example 3.8.

The cup product formula $(\varphi \smile \psi)(\sigma) = \varphi(\sigma \,|\, [v_0,\cdots,v_k])\psi(\sigma \,|\, [v_k,\cdots,v_{k+\ell}])$ also gives relative cup products

$$H^k(X;R) \times H^\ell(X,A;R) \xrightarrow{\ \smile\ } H^{k+\ell}(X,A;R)$$
$$H^k(X,A;R) \times H^\ell(X;R) \xrightarrow{\ \smile\ } H^{k+\ell}(X,A;R)$$
$$H^k(X,A;R) \times H^\ell(X,A;R) \xrightarrow{\ \smile\ } H^{k+\ell}(X,A;R)$$

since if φ or ψ vanishes on chains in A then so does $\varphi \smile \psi$. There is a more general relative cup product

$$H^k(X,A;R) \times H^\ell(X,B;R) \xrightarrow{\ \smile\ } H^{k+\ell}(X,A\cup B;R)$$

when A and B are open subsets of X or subcomplexes of the CW complex X. This is obtained in the following way. The absolute cup product restricts to a cup product $C^k(X,A;R)\times C^\ell(X,B;R) \to C^{k+\ell}(X,A+B;R)$ where $C^n(X,A+B;R)$ is the subgroup of $C^n(X;R)$ consisting of cochains vanishing on sums of chains in A and chains in B. If A and B are open in X, the inclusions $C^n(X,A\cup B;R) \hookrightarrow C^n(X,A+B;R)$ induce isomorphisms on cohomology, via the five-lemma and the fact that the restriction maps $C^n(A\cup B;R) \to C^n(A+B;R)$ induce isomorphisms on cohomology as we saw in the discussion of excision in the previous section. Therefore the cup product $C^k(X,A;R)\times C^\ell(X,B;R) \to C^{k+\ell}(X,A+B;R)$ induces the desired relative cup product

$H^k(X,A;R)\times H^\ell(X,B;R)\to H^{k+\ell}(X,A\cup B;R)$. This holds also if X is a CW complex with A and B subcomplexes since here again the maps $C^n(A\cup B;R)\to C^n(A+B;R)$ induce isomorphisms on cohomology, as we saw for homology in §2.2.

Proposition 3.10. *For a map $f:X\to Y$, the induced maps $f^*:H^n(Y;R)\to H^n(X;R)$ satisfy $f^*(\alpha\smile\beta)=f^*(\alpha)\smile f^*(\beta)$, and similarly in the relative case.*

Proof: This comes from the cochain formula $f^\sharp(\varphi)\smile f^\sharp(\psi)=f^\sharp(\varphi\smile\psi)$:

$$\begin{aligned}(f^\sharp\varphi\smile f^\sharp\psi)(\sigma)&=f^\sharp\varphi(\sigma|[v_0,\cdots,v_k])f^\sharp\psi(\sigma|[v_k,\cdots,v_{k+\ell}])\\&=\varphi(f\sigma|[v_0,\cdots,v_k])\psi(f\sigma|[v_k,\cdots,v_{k+\ell}])\\&=(\varphi\smile\psi)(f\sigma)=f^\sharp(\varphi\smile\psi)(\sigma)\end{aligned}$$ ☐

We now define the **cross product** or **external cup product**. The absolute and general relative forms are the maps

$$H^k(X;R)\times H^\ell(Y;R)\xrightarrow{\times}H^{k+\ell}(X\times Y;R)$$
$$H^k(X,A;R)\times H^\ell(Y,B;R)\xrightarrow{\times}H^{k+\ell}(X\times Y,A\times Y\cup X\times B;R)$$

given by $a\times b=p_1^*(a)\smile p_2^*(b)$ where p_1 and p_2 are the projections of $X\times Y$ onto X and Y.

Example 3.11: The *n*-Torus. For the n-dimensional torus T^n, the product of n circles, let us show that all cohomology classes are cup products of 1-dimensional classes. More precisely, we show that $H^k(T^n;R)$ is a free R-module with basis the cup products $\alpha_{i_1}\smile\cdots\smile\alpha_{i_k}$ for $i_1<\cdots<i_k$, where $\alpha_i\in H^1(T^n;R)$ is $p_i^*(\alpha)$ for α a generator $H^1(S^1;R)$ and p_i the projection of T^n onto its i^{th} factor.

As a preliminary step we show that for α a generator of $H^1(I,\partial I;R)$, the map

$$H^n(Y;R)\to H^{n+1}(I\times Y,\partial I\times Y;R),\qquad \beta\mapsto\alpha\times\beta$$

is an isomorphism for all spaces Y. This uses commutativity of the following square:

$$\begin{array}{ccc}H^k(A;R)\times H^\ell(Y;R) & \xrightarrow{\delta\times 1\!\!1} & H^{k+1}(X,A;R)\times H^\ell(Y;R)\\ \downarrow\times & & \downarrow\times\\ H^{k+\ell}(A\times Y;R) & \xrightarrow{\delta} & H^{k+\ell+1}(X\times Y,A\times Y;R)\end{array}$$

To check this, start with an element of the upper left product, represented by cocycles $\varphi\in C^k(A;R)$ and $\psi\in C^\ell(Y;R)$. Extend φ to a cochain $\overline{\varphi}\in C^k(X;R)$. Then the pair (φ,ψ) maps rightward to $(\delta\overline{\varphi},\psi)$ and then downward to $p_1^\sharp(\delta\overline{\varphi})\smile p_2^\sharp(\psi)$. Going the other way around the square, (φ,ψ) maps downward to $p_1^\sharp(\varphi)\smile p_2^\sharp(\psi)$ and then rightward to $\delta(p_1^\sharp(\overline{\varphi})\smile p_2^\sharp(\psi))$ since $p_1^\sharp(\overline{\varphi})\smile p_2^\sharp(\psi)$ extends $p_1^\sharp(\varphi)\smile p_2^\sharp(\psi)$ over $X\times Y$. Finally, $\delta(p_1^\sharp(\overline{\varphi})\smile p_2^\sharp(\psi))=p_1^\sharp(\delta\overline{\varphi})\smile p_2^\sharp(\psi)$ since $\delta\psi=0$.

Returning to the product $I\times Y$, the long exact sequence for the pair $(I\times Y,\partial I\times Y)$ breaks up into split short exact sequences

$$0\longrightarrow H^n(I\times Y,R)\longrightarrow H^n(\partial I\times Y;R)\xrightarrow{\delta}H^{n+1}(I\times Y,\partial I\times Y;R)\longrightarrow 0$$

The map δ is an isomorphism when restricted to the copy of $H^n(Y;R)$ corresponding to $\{0\}\times Y$. This copy of $H^n(Y;R)$ consists of elements of the form $1_0\times\beta$ where $1_0\in H^0(\partial I;R)$ is represented by the cocycle that is 1 on $0\in\partial I$ and 0 on $1\in\partial I$. By the commutative square above, $\delta(1_0\times\beta)=\delta(1_0)\times\beta$. The element $\delta(1_0)$ is a generator of $H^1(I,\partial I;R)$, by the case that Y is a point. Any other generator α is a scalar multiple of $\delta(1_0)$ by a unit of R, so this shows the map $\beta\mapsto\alpha\times\beta$ is an isomorphism.

An equivalent statement is that the map $H^n(Y;R)\to H^{n+1}(S^1\times Y,\{s_0\}\times Y;R)$, $\beta\mapsto\alpha\times\beta$, is an isomorphism, with α now a generator of $H^1(S^1,s_0;R)$. Via the long exact sequence of the pair $(S^1\times Y,\{s_0\}\times Y)$, this implies that the map

$$H^{n+1}(Y;R)\times H^n(Y;R)\to H^{n+1}(S^1\times Y;R),\qquad(\beta_1,\beta_2)\mapsto 1\times\beta_1+\alpha\times\beta_2$$

is an isomorphism, with α a generator of $H^1(S^1;R)$. Specializing to the case of the n-torus, we conclude by induction on n that $H^k(T^n;R)$ has the structure described at the beginning of the example.

We can use this calculation to deduce a fact that will be used shortly in the calculation of cup products in projective spaces. Writing $n=i+j$, the cube I^n is the product $I^i\times I^j$, and the assertion is that the cross product of generators of $H^i(I^i,\partial I^i;R)$ and $H^j(I^j,\partial I^j;R)$ is a generator of $H^n(I^n,\partial I^n;R)$, where we are using the first of the following three cross products:

$$\begin{aligned} H^i(I^i,\partial I^i;R)\times H^j(I^j,\partial I^j;R)&\xrightarrow{\ \times\ }H^n(I^n,\partial I^n;R)\\ H^i(T^i,\dot T^i;R)\times H^j(T^j,\dot T^j;R)&\xrightarrow{\ \times\ }H^n(T^n,\dot T^n;R)\\ H^i(T^i;R)\times H^j(T^j;R)&\xrightarrow{\ \times\ }H^n(T^n;R)\end{aligned}$$

In the second cross product, the dots denote deletion of the top-dimensional cell. All three cross products are equivalent. This is evident for the first two, thinking of the torus as a quotient of a cube. For the second two, note that all cellular boundary maps for T^n with $\mathbb{Z}$ coefficients must be trivial, otherwise the cohomology groups would be smaller than computed above. Hence all cellular coboundary maps with arbitrary coefficients are zero, and the map $H^n(T^n,\dot T^n;R)\to H^n(T^n;R)$ is an isomorphism. The corresponding results for T^i and T^j are of course true as well.

Since cross product is associative, the earlier calculation shows that for the last of the three cross products above, the cross product of generators is a generator, so this is also true for the first cross product.

The Cohomology Ring

Since cup product is associative and distributive, it is natural to try to make it the multiplication in a ring structure on the cohomology groups of a space X. This is easy to do if we simply define $H^*(X;R)$ to be the direct sum of the groups $H^n(X;R)$. Elements of $H^*(X;R)$ are finite sums $\sum_i\alpha_i$ with $\alpha_i\in H^i(X;R)$, and the product of